



Prop. Normal and Nilpotent $\Rightarrow$ Zero.

Proof. Let $T: V \rightarrow V$ be a linear operator on fin. dim inn. prod space V . Since T is normal, there exists a basis $\beta = \{\alpha_1, \dots, \alpha_n\}$ for V consisted by eigenvectors of T . Given $\alpha \in V$,

$$\alpha = x_1 \alpha_1 + \dots + x_n \alpha_n \quad \text{for some } x_i \in F$$

and so $T\alpha = x_1 C_1 \alpha_1 + \dots + x_n C_n \alpha_n$, where C_i are eigenvalues corresponding to α_i . Meanwhile, since T is nilpotent, all eigenvalues for T are zero, so $T\alpha = 0$. Thus, $T = 0$.

Alternatively, without spectral decomposition, observe that: if T is normal,

$$\|T\alpha\|^2 = \langle T\alpha, T\alpha \rangle = \langle \alpha, T^* T \alpha \rangle = \langle \alpha, T T^* \alpha \rangle = \langle T^* \alpha, T^* \alpha \rangle = \|T^* \alpha\|^2$$

so $T\alpha = 0 \Leftrightarrow T^* \alpha = 0$, which gives $\ker T = \ker T^*$.

Now, suppose that T is nilpotent and $T \neq 0$. Take $m \in \mathbb{N}$ be the smallest integer such that $T^m = 0$. (That is, $T^0 \dots, T^{m-1} \neq 0$).

Let $\alpha \in V$ such that $T^{m-1} \alpha \neq 0$. Then, since $T^m = 0$, $T^{m-1} \alpha \in \ker T$.

Since $\ker T = \ker T^*$, $T^* T^{m-1} \alpha = 0$. So,

$$\langle T^* T^{m-1} \alpha, \dots \rangle$$

Prop.

Let T be a normal operator on a fin. dim complex inn. prod space V .

Then, T^* is polynomial in T .

Proof. Since T is normal, there is a spectrum decomposition

$$T = c_1 T_1 + \dots + c_k T_k$$

where c_i are disjoint eigenvalues of T and

T_i are orthogonal projection onto $\ker(T - c_i I)$.

$$\text{So, } T^* = \overline{c_1} T_1^* + \dots + \overline{c_k} T_k^* = \overline{c_1} T_1 + \dots + \overline{c_k} T_k,$$

Meanwhile, since each T_i 's are orthogonal, given polynomial $f(t) \in \mathbb{C}[t]$,

$$f(T) = f(c_1) T_1 + \dots + f(c_k) T_k,$$

therefore let

$$f = \sum_{i=1}^k \overline{c_i} \prod_{j \neq i} \frac{(x - c_j)}{(c_i - c_j)}$$

Then, f is $\mathbb{C}$ -polynomial and

$$f(c_i) = \overline{c_i}$$

So that

$$f(T) = T^*, \text{ as we desired. } \square$$